Department of Electrical and Electronic Engineering

EEE499P :Final Year Design Project (FYDP)

Weekly Update

Week Number: 02

Date: From 19 Oct 2025 To 26Oct 2025.

<u>ATC Panel:</u>

Chair:Dr. Mohammed Belal Hossain Bhuian

Md. Mahmudul Islam

Sabbir Ahmed

<u>FYDP Group Members:</u>

Shahriar Ahmed

Aban Ahmad

Maisha Mahjabin Nidhi

Daiyan Ibnul Aswad

<u>Week01 Update on:</u> Identification of a real life engineering problem for the proposal Incorporation of 2 new topics and description of engineer meeting.

<u>Topic 1:</u> A single line-to-ground (LG) fault detection of 33 kV line under BREB.

<u>Consultation with faculty about fault detection in transmission lines:</u>

Dr.Shameem Ahmad

Associate Professor,BSRM School of Engineering.

Output Summary:

Sir suggested us to go for the distribution lines of urban areas, as transmission lines are designed with stronger structures, higher clearances, better insulation and controlled right of way which reduces the chance of faults. On the other hand, distribution lines are located close to consumers and ground level which makes them highly exposed to environmental factors and physical damage from vehicles or construction works. These lines also experience higher load density and frequent network modifications, which can lead to overloading, weak joints, and insulation failure.

1. **Overloading effect/ thermal limit is the most important problem**
2. **Sag identification**
3. **Wire Too close to residential houses**

<u>Consultation with Engineers about fault detection in distribution lines</u>:

1.Engr. Al Mamun

CEO,Controlware Engineering Limited.

2.Golam Mawla

Sub Divisional Engineer,Desco Bangladesh.

<u>Outcome Summary:</u>

In urban areas like Dhaka, the 11 kV distribution network under DPDC/DESCO is densely located within a small region. The substations, feeders, switching devices, and protection systems are installed close to each other, so any abnormal condition like a line insulation failure or earth fault is detected very quickly by protection relays and monitored efficiently through SCADA and automated switching systems. Also, because the lines are shorter and well-planned, maintenance teams can reach the fault spot faster, reducing the likelihood of faults escalating or recurring.

On the other hand, BREB's 33 kV overhead feeders extend across large rural distances and cover villages, fields, rivers, and forested areas. These long lines are more exposed to weather changes and physical obstacles, making faults more likely. Due to the wide geographical spread, patrolling and locating the exact fault point takes more time. Limited automation and fewer monitoring devices mean faults may remain undetected or unattended for longer, which increases the overall observed number of outages and faults.

<u>Problem: Detecting and distinguishing real faults in high-current overhead power lines.</u>

<u>Topic 2: Purification or Removal of Ultra-Fine Particles caused by Air Pollution</u>

Problem:

Air Pollution.

Conventional mechanical filters and standard Electrostatic Precipitators (ESPs) show significant **limitations in capturing ultrafine and nanosized particles**, primarily due to low charging efficiency, particle re-entrainment and high energy or maintenance costs.

The removal of ultra-fine particles such as PM (particulate matter), fine dust particles and nanoparticles (<100nm) remains a challenge.

Although there are several methods of gas ionization, such as triboelectric, ultraviolet, and radiation effects, for applications in industrial precipitators, corona discharge is universally used as the most efficient and economical approach

Methods:

1) **Electrostatic Precipitators**

Key Design Features:

Corona charging electrode: Tungsten or stainless-steel wire (HV: 10–15 kV).

Collector plate: Aluminum or copper plates spaced 20–40 mm apart.

Air duct: Transparent acrylic chamber for visualization and particle measurement.

Power supply: Variable high-voltage DC converter with current limiting.

Instrumentation: Ozone sensor, aerosol counter (CPC/OPC), pressure and flow sensors.

Expected Advantages:

Continuous operation and self-cleaning (with proper polarity reversal).

Effective for both fine and coarse particles.

Expected Challenges:

Ozone generation from corona discharge.

Reduced efficiency for nanoparticles (<50 nm).

High-voltage safety concerns and maintenance

2) **Electret**

Key Design Features:

Material: PVDF or polypropylene (PP) nanofiber mats fabricated via electrospinning.

Charging method: Corona charging or thermal poling to induce stable dipoles.

Structure: Multi-layer nanofiber mat mounted in a cylindrical or planar frame.

Instrumentation: Surface potential meter, particle counter, and airflow sensors.

Expected Advantages:

Low energy consumption (no continuous HV).

No ozone generation.

Excellent capture of ultrafine and neutral particles.

Potential for hybridization with mechanical pre-filters.

Expected Challenges:

Charge decay under humidity and temperature variations.

Limited lifespan or regeneration cycles.

Mechanical fragility of nanofibers.